CLINICAL GENOMICS & PHARMACOGENOMICS ACTION BRIEF

Patient: Daniel Ehrle (Daniel_Ehrle) · **Date:** October 13, 2026 · **Primary Audience:** Patient, Primary Care Physician (PCP) & Specialists

SECTION 1 · PRIORITY GENOMIC FINDINGS, CARRIER STATUS & CLINICAL MEANING

GENE & BIOLOGICAL SYSTEM	VARIANT & ZYGOSITY	CLASSIFICATION & CONDITION	CLINICAL IMPACT & PRACTICAL GUIDANCE
CBLIF **CBLIF** Multisystem	**c.79+1G>A · rs147785187** **Heterozygous** Autosomal Recessive	**Pathogenic** Intrinsic Factor Deficiency / Cobalamin Malabsorption	Single-copy heterozygous carrier preserves intrinsic factor production. Check periodic serum B12 during routine wellness visits.
GJB2 **GJB2** Multisystem	**p.Met34Thr · c.101T>C · rs35887622** **Heterozygous** Autosomal Recessive, Autosomal Dominant	**Pathogenic** Autosomal Recessive Deafness 1A / Dominant 3A	Single-copy heterozygous carrier maintains normal adult hearing. Consider baseline hearing check; avoid unnecessary ototoxic aminoglycosides.
VDR **VDR** Multisystem	**c.-1172A>G · rs4516035** **Homozygous** Pathway Modifier	**Likely pathogenic** Chronic obstructive pulmonary disease	Modulates calcitriol receptor signaling. Check routine 25(OH)D at annual checkup to guide standard vitamin D3 supplementation.
F5 **F5** Multisystem	**p.Arg534Gln · c.1601G>A · rs6025** **Heterozygous** Autosomal recessive / AD	**drug response** Congenital factor V deficiency	Avoid systemic oral estrogen therapy (thromboembolism risk). Implement standard perioperative clot prophylaxis during major surgery.
PEX6 **PEX6** Heart & Cardiovascular	**p.Val788Met · c.2362G>A · rs267608240** **Heterozygous** Autosomal recessive	**Conflicting classifications of pathogenicity** Peroxisome biogenesis disorder 4A (Zellweger)	Single-copy heterozygous carrier preserves normal peroxisomal function. Standard annual liver and lipid wellness panels.
GJB3 **GJB3** Heart & Cardiovascular	**p.Arg75Cys · c.223C>T · rs370476720** **Heterozygous** Autosomal recessive	**Conflicting classifications of pathogenicity** Inborn genetic diseases	Single-copy heterozygous carrier maintains normal adult hearing. Consider baseline hearing check; avoid unnecessary ototoxic aminoglycosides.
SCN4A **SCN4A** Heart & Cardiovascular	**p.His599Arg · c.1796A>G · rs187401185** **Heterozygous** Autosomal dominant / AR	**Conflicting classifications of pathogenicity** Paramyotonia congenita of Von Eulenburg	Maintain normal electrolyte balance. Inform anesthesiology regarding depolarizing muscle relaxants prior to general surgery.
C19orf12 **C19orf12** Multisystem	**p.Gly42Arg · c.124G>A · rs200133991** **Heterozygous** Autosomal Recessive	**Conflicting classifications of pathogenicity** Neurodegeneration with Brain Iron Accumulation 4	Single-copy heterozygous finding; review in the context of standard annual wellness labs and personal clinical history.
LRP5 **LRP5** Heart & Cardiovascular	**p.Val233Glu · c.698T>A · rs1177484229** **Heterozygous** Pathway Modifier	**Uncertain significance** Retinal dystrophy	Modulates lipoprotein clearance. Review routine fasting lipid panel and cardiovascular wellness goals at annual checkup.
BLM **BLM** Multisystem	**p.Val724Ala · c.2171T>C · rs750687788** **Heterozygous** Autosomal recessive	**Uncertain significance** Hereditary cancer-predisposing syndrome	Moderately elevated cancer susceptibility. Discuss guideline-directed breast/preventive screening (annual MRI/mammogram) with PCP.
ANK2 **ANK2** Multisystem	**p.Arg3906Trp · c.11716C>T · rs121912706** **Heterozygous** Pathway Modifier	**Conflicting classifications of pathogenicity** ANK2-related disorder	Modulates cardiac repolarization. Consider baseline ECG at routine physical; review QT-prolonging medications.
NOD2 **NOD2** Multisystem	**p.Arg675Trp · c.2023C>T · rs2066844** **Heterozygous** Pathway Modifier	**Conflicting classifications of pathogenicity|association** Regional enteritis	Single-copy heterozygous finding; review in the context of standard annual wellness labs and personal clinical history.

SECTION 2 · HIGH-PRIORITY MEDICATION & PHARMACOGENOMIC PRECAUTIONS (CPIC / FDA)

GENE & VARIANT / DIPLOTYPE	PHENOTYPE / STATUS	AFFECTED MEDICATION(S)	PRESCRIBING ACTION & CLINICAL GUIDANCE
F5 · p.Arg534Gln · c.1601G>A · rs6025 Heterozygous APC Resistance	**Heterozygous APC Resistance**	⚠️ **HIGH RISK · AVOID** **Systemic Estrogens, Combined Oral Contraceptives, HRT**	**AVOID SUBSTRATE:** Confers Activated Protein C (APC) resistance with 35-fold amplified risk of venous thromboembolism (VTE) and pulmonary embolism with systemic estrogen exposure. Avoid oral estrogen formulations; implement situational perioperative VTE prophylaxis.
CYP2C9 · *1/*2 R144C (rs1799853) C>T	**Intermediate Metabolizer**	⬇️ **HIGH RISK · REDUCE DOSE** **Celecoxib (Celebrex), Phenytoin (Dilantin), Warfarin (Coumadin)**	**REDUCE STARTING DOSE:** Moderate decrease in celecoxib clearance. Initiate therapy at lowest recommended dose and monitor for GI/cardiovascular adverse effects. Initiate at reduced starting dose per CPIC.
CYP2C19 · *1/*17 I331V (rs3758581) A>G, expression (rs12248560) C>T	**Rapid Metabolizer**	⬇️ **HIGH RISK · REDUCE DOSE** **Voriconazole**	**REDUCE STARTING DOSE:** Increased clearance and lower trough concentrations. Perform therapeutic drug monitoring (TDM) to avoid subtherapeutic levels. Initiate at reduced starting dose per CPIC.
CYP3A5 · *3/*3 splice defect (rs776746) T>C	**Poor Metabolizer**	⬇️ **HIGH RISK · REDUCE DOSE** **Ondansetron**	**REDUCE STARTING DOSE:** Genotype CC is associated with decreased metabolism of ondansetron as compared to genotypes CT + TT. Initiate at reduced starting dose per CPIC.
UGT1A1 · *1/*28_80 5'UTR (rs887829) C>T, Promoter (rs3064744) insTA	**Intermediate Metabolizer**	**CAUTION / MONITOR** **Belinostat, Irinotecan (Camptosar)**	**CAUTION / MONITOR:** Monitor for increased hematologic toxicity. Monitor clinical response and consider dose adjustment per CPIC.
NAT2 · *5/*5 I114T (rs1801280) T>C	**Poor Metabolizer**	**CAUTION / MONITOR** **Isoniazid**	**CAUTION / MONITOR:** NAT2 *5/*5 slow acetylator phenotype. Higher risk of isoniazid-induced hepatotoxicity and peripheral neuropathy. Monitor clinical response and consider dose adjustment per CPIC.

PREVENTIVE CARE, CLINICAL PERSPECTIVE & EHR SUMMARY

Patient: Daniel Ehrle (`Daniel_Ehrle`) · Annual Checkup Topics, Clinical Perspective & Physician Handoff Note

SECTION 3 · HIGH-LEVEL PREVENTIVE CARE & ANNUAL CHECKUP DISCUSSION POINTS

GENOMIC FINDING	BIOLOGICAL PATHWAY	PREVENTIVE CARE TOPIC	CARE SETTING & TIMING	CLINICAL CONTEXT & GUIDANCE
CBLIF `c.79+1G>A · rs147785187` **Heterozygous (Autosomal Recessive)**	**CBLIF · Vitamin B12 (Cobalamin) & Folate Absorption** Pathogenic	**Periodic Vitamin B12 & Folate Check** Routine Wellness Lab Panel	**Primary Care Physician (PCP)** **Routine Annual or Biennial Checkup**	Single-copy heterozygous carrier preserves intrinsic factor production. Check periodic serum B12 during routine wellness visits.
GJB2 `p.Met34Thr · c.101T>C · rs35887622` **Heterozygous (Autosomal Recessive, Autosomal Dominant)**	**GJB2 · Cochlear Gap Junction & Hearing Health** Pathogenic	**Baseline Hearing Check & Ototoxic Medication Precaution** Routine Preventive Care	**Primary Care Physician (PCP)** & **Audiology** **Baseline check; repeat if symptoms arise**	Single-copy heterozygous carrier maintains normal adult hearing. Consider baseline hearing check; avoid unnecessary ototoxic aminoglycosides.
VDR `c.-1172A>G · rs4516035` **Homozygous (Pathway Modifier)**	**VDR · Vitamin D Signaling & Bone Health** Likely pathogenic	**Routine Vitamin D & Calcium Check** Annual Wellness Panel	**Primary Care Physician (PCP)** **Annual PCP Checkup**	Modulates calcitriol receptor signaling. Check routine 25(OH)D at annual checkup to guide standard vitamin D3 supplementation.
F5 `p.Arg534Gln · c.1601G>A · rs6025` **Heterozygous (Autosomal recessive / AD)**	**F5 · Blood Coagulation & Clot Risk Management** drug response	**Situational Clot Prevention (Surgery, Immobility, Estrogen Avoidance)** Perioperative & Prescribing Precaution	**Primary Care Physician (PCP)** **Situational (Before surgery, long travel, or hormone therapy)**	Avoid systemic oral estrogen therapy (thromboembolism risk). Implement standard perioperative clot prophylaxis during major surgery.
PEX6 `p.Val788Met · c.2362G>A · rs267608240` **Heterozygous (Autosomal recessive)**	**PEX6 · Peroxisomal & Lipid Metabolism** Conflicting classifications of pathogenicity	**Routine Liver & Lipid Wellness Panel** Standard Annual Physical Bloodwork	**Primary Care Physician (PCP)** **Annual PCP Checkup**	Single-copy heterozygous carrier preserves normal peroxisomal function. Standard annual liver and lipid wellness panels.

SECTION 4 · CLINICAL PERSPECTIVE: BALANCING PROACTIVE CARE VS. OVER-INTERVENTION

✅ Why These Findings Are Useful

- **Proactive Medication Safety:** Recording pharmacogenomic metabolizer status in the medical chart helps select effective medications and avoid preventable side effects before a prescription is started.
- **Informed Annual Checkups:** Highlighting relevant biological pathways allows the Primary Care Physician to tailor routine wellness checks and age-appropriate preventive screening.

🛑 Why Aggressive Intervention Is Not Needed

- **Single-Copy Carrier Reassurance:** No two-copy (biallelic) recessive disease genotypes are present; single-copy recessive carriers retain a functional gene copy and maintain normal baseline health.
- **Modifier Variants Are Not Diagnoses:** Common pathway modifiers indicate mild physiological tendencies and should never trigger unnecessary procedures or unprovoked medication changes.

SECTION 5 · COPY-PASTE PHYSICIAN HANDOFF / EHR CHART SUMMARY

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CLINICAL GENOMICS & PHARMACOGENOMICS SUMMARY (Daniel Ehrle | ID: Daniel_Ehrle)
• PRIORITY GENOMIC & CARRIER FINDINGS: CBLIF (c.79+1G>A · rs147785187, Heterozygous), GJB2 (p.Met34Thr · c.101T>C · rs35887622, Heterozygous), VDR (c.-1172A>G · rs4516035, Homozygous), F5 (p.Arg534Gln · c.1601G>A · rs6025, Heterozygous), PEX6 (p.Val788Met · c.2362G>A · rs267608240, Heterozygous), GJB3 (p.Arg75Cys · c.223C>T · rs370476720, Heterozygous). Single-copy (heterozygous) recessive carrier states retain one functional gene copy and do not cause primary recessive disease.
• MEDICATION & EXPOSURE PRECAUTIONS (CPIC / FDA):
  - AVOID / CONTRAINDICATED: Systemic Estrogens, Combined Oral Contraceptives, HRT
  - REDUCE STARTING DOSE: Celecoxib (Celebrex), Phenytoin (Dilantin), Warfarin (Coumadin) · Voriconazole · Ondansetron
  - RAPID METABOLISM / MONITOR: Belinostat, Irinotecan (Camptosar) · Isoniazid
• PREVENTIVE CARE PLAN: Review high-level wellness and screening topics in Section 3 during routine primary care checkups.
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